

Universal PWM SIGNAL ISOLATION MODULE

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This PWM (pulse width modulation) signal isolation module is built around the widely adopted 6N137 high-speed optocoupler. The device combines a high-efficiency input LED with an integrated optical photodiode detector IC to ensure robust signal isolation.

The advanced 6N137 variants incorporate an open drain NMOS transistor output (photodetector logic gate), offering lower leakage than traditional open-collector Schottky-clamped transistor outputs (VO). An integrated enable function (VE) also allows the detector to be strobed, improving design flexibility and control.

Optocoupler speed is critical for its performance. According to

the Everlight 6N137 datasheet, maximum output rise and fall times are 40ns and 10ns, respectively. These represent the output voltage transition times when the input LED is pulsed with forward current. The inverse of the sum of these times closely approximates the maximum operating frequency under specified drive conditions.

Fig. 1 shows how the 6N137 IC module looks and Fig. 2 shows its internal structure.

PWM resolution defines the number of distinct duty cycles within a given period. A 10-bit resolution yields 1024 steps, while an 8-bit resolution provides 256 discrete duty cycles.

Circuit and working

Fig. 3 shows the circuit diagram of the PWM signal isolation module. The 100nF bypass capacitor (C1) is a crucial component. It can be either a ceramic or solid tantalum capacitor with good high-frequency characteristics; it should be connected as close as possible to the VCC and GND pins of the 6N137 (IC1). No pull-up resistor is required for the enable pin (VE), as the chip has

TABLE 1
TRUTH TABLE OF IC 6N137

Input (Pin2)	Enable (Pin7)	Output (Pin6)
H(5V)	H(5V)	L(0V)
L(0V)	H(5V)	H(5V)
H(5V)	L(0V)	H(5V)
L(0V)	L(0V)	H(5V)
H(5V)	NC	L(0V)
L(0V)	NC	H(5V)

PARTS LIST

Semiconductors:

- IC1 - 6N137 high-speed optocoupler
LED1 - 3mm LED (optional)

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 330 Ω
R2 - 4.7k Ω
R3 - 10k Ω
R4 - 1k Ω (optional)

Capacitors:

- C1 - 10nF ceramic disk

Miscellaneous:

- CON1-CON3 - 2-pin connector
- 5V DC supply
- Wires and jumpers



Fig. 1: 6N137 high-speed optocoupler isolation module

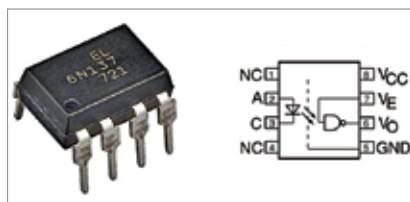


Fig. 2: 6N137 IC and its internal circuit

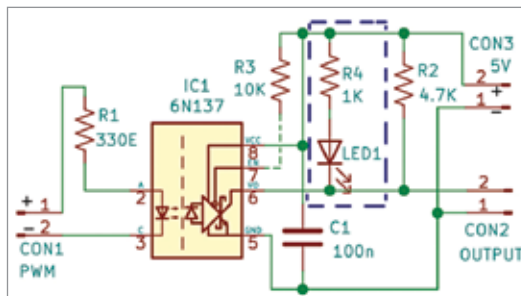


Fig. 3: Circuit diagram

an internal pull-up. Therefore, the external pull-up resistor (R3 shown in dotted line) is not necessary.

The output (VO) of the 6N137 is an open drain transistor, and hence it can either sink current or remain

floating. To avoid a floating output, a pull-up resistor (R2) is employed. Since the output is of the open-drain type, when the input is high, the output will be low, and vice versa. The typical operating voltage is 5V. The truth table of the 6N137 with respect to GND is shown in Table 1.